

**A Commentary on the Differential Development of Face- and Body-Selective Regions in the
Fusiform Gyrus**

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Introduction

As humans, our day-to-day activities involve interacting with others and perceiving them. This includes recognizing and understanding facial expressions, actions, and body movements in order to gauge the objective of the interaction. Whether it be calculating the next move of your tennis opponent or debating a topic with a friend, perception of facial expressions and the body is an essential factor. To understand the mechanisms behind these crucial cognitive processes, specific regions within the visual cortex have been pinpointed to contribute to face and body perception thanks to recent advancements in neuroimaging.

Background and contextual information

Through the use of functional magnetic resonance imaging (fMRI), it has been found that the fusiform face area (FFA) and the fusiform body area (FBA), located in the right fusiform gyrus, and the extrastriate body area (EBA), located posterior to the FFA and FBA, are active when looking at faces and bodies (Kanwisher et al., 1997; Downing et al., 2001). The fMRI is the preferred method for imaging brain areas and their activation due to its high spatial resolution that allows us to follow the oxygenation level-dependent (BOLD) signals that occur during neuronal activation. Within the complex visual cortex, the fusiform gyrus is located at the lower surface of the occipital and temporal lobes and plays a significant role in visual processing (Weiner & Zilles, 2016). It is the largest structure and has many different functions including object recognition, face and body perception, and processing visual stimuli, among others. Within the fusiform gyrus, the FFA has been identified to be the key region in processing facial identity (Grill-Spector et al., 2004). Previous studies have established FFA activation in both facial detection and identification through image identification tasks and analyzing BOLD signals provided by fMRI. However, when past participants were presented with images of non-face objects, the FFA did not show any activation. The FBA lies within the lateral posterior fusiform gyrus and has been found to be highly sensitive to perceiving the human body (Peelen et al., 2005). In a study conducted by Taylor et al. (2007), body selectivity was determined through the increase of activation in the FBA when viewing

images of bodies, but not the FFA. The EBA is another region involved in body selectivity, with strong activation in response to bodies but not faces, and is located in the posterior inferior temporal sulcus (Downing et al., 2001). Focusing on the face- and body-selective areas in the fusiform gyrus, although Peelen et al. (2005) found that these two areas have almost identical anatomical overlap, Schwarzlose et al., (2005) discovered that the FFA and FBA are two separate entities with the aid of the high-resolution fMRI that has the ability to distinguish between the two areas. To consider the FFA and FBA as separate entities, then the development of either area should also be considered independently. Previous fMRI studies and face recognition tasks in children have found that the FFA increases in size at an extremely slow pace, growing from childhood and throughout adolescence (Golarai et al., 2010), along with a positive correlation in performance on face perception tasks (Golarai et al., 2007). However, not much was known about the developmental trajectory of the FBA or the EBA, until the investigation carried out by Peelen et al. (2009) revealed the similarities and differences.

Summary of the original article

Peelen et al.'s 2009 study explores the developmental trajectories of selectivity for faces and bodies in the fusiform gyrus. The experiments focused on determining whether the proximity and overlap of specific brain areas influenced their growth paths.

Methods

44 participants, ranging from 7-32 years old and with no known medical history, were involved in the experiment. 22 participants were between the ages of 7 and 17, and 22 participants were 18 years old and up. The former group, although including adolescents, was referred to as "children," while the latter group was referred to as adults. The responses of one adult participant could not be recorded.

Using fMRI, participants' face- and body-selective areas were scanned while they were presented with 21 15-second image blocks. While 5 of the image blocks were fixation-only, baseline conditions, 16 image blocks involved images of faces, headless bodies, tools, or various scenes. During each image block, a one-back repetition task was carried out in order to keep participants engaged, where a couple of the images were displayed twice and the participants were asked to detect the image repetition.

Analyses on the regions of interest (ROI) mainly focused on the right FFA (rFFA), right FBA (rFBA), and right EBA (rEBA) due to the differing roles of the left fusiform gyrus and the right fusiform gyrus, and the lack of activation in the left hemisphere. The rFFA is highly active by facial parts, while the rFBA is activated by the complete processing of faces (Rossion et al., 2000). To determine the overlap between the rFFA and rFBA, an index was calculated by dividing the number of overlapping voxels by the number of voxels of the smaller of the two ROIs. An index 1 means that the smaller of the two ROIs lies completely within the other, with a 100% overlap. An index of 0 means that there is no overlap. Additionally, whole-brain analyses were conducted in the faces versus tools condition and the bodies versus two conditions separately for children and adults.

Results

A two-way ANOVA examining the effects of age (children vs. adults) and category (faces vs. bodies) was utilized to observe the interaction between age and category in facial and body processing. It was found that adults were more accurate than children during tasks, but there was no category influence on performance. There was no significant interaction between age and category. Reaction times were investigated, showing that repetition in face images was found faster compared to body images. Furthermore, age was not a significant predictor in reaction times and there was no significant interaction between age and category.

To assess the developmental trajectories of the rFFA, rFBA, left EBA (lEBA), and right EBA (rEBA), analysis was conducted at three statistical thresholds ($p < .05$, $p < .01$, $p < .001$). Talairach coordinates report the locations of activation spots in brain regions on fMRI (Lancaster et al., 2007), and it was found that the average Talairach coordinates of the peak activation for the rFFA and rFBA were nearly indistinguishable, suggesting the two regions lie in the same space of the visual cortex. No difference in the amount of overlap between the rFFA and rFBA was found in children versus adults, with a 0.75 overlap index.

While the size of the rFFA increased with age, there was no correlation between the size of the rFBA and age. The rFBA was significantly larger than the rFFA in children, around 70% larger, while for adults, there were no significant differences between the sizes of the rFBA and the rFFA. This serves as evidence that as the rFFA grows over time, the rFBA is consistent in size since childhood.

A follow-up study conducted to determine if the increase in the size of the rFFA between the children group and adult group is gradual or not found that there was a gradual development of the rFFA that continues during adolescence. The children group was divided into two subgroups: child (7-11 years old) and adolescent (12-17 years old). At all thresholds, the rFFA size was significantly smaller in the children group than in the adult group, but a significant difference was only found between the child and adolescent group at one threshold ($p < .05$). No significant differences were found between the adolescent and adult group. There were no significant correlations found between the size of the rFFA and accuracy or response times for any of the three thresholds.

The degree of selectivity of the ROIs was determined by analyzing the peak T-value of each. The peak T-value of the rFFA gradually increased with age, but there was no significant correlation between the peak T-value of the rFBA and age. When comparing within each age group, the average peak T-value of the rFFA was significantly higher than the rFBA in the adult group, versus the comparable peak T-values

of the rFFA and rFBA in the children group. Furthermore, the peak T-value of the rFFA was lowest for the child group and highest for the adult group, and intermediate for the adolescent group. No significant difference was found between the adolescent group with either the children or adult groups. To add on, no significant correlations were found between the age-related increase of the peak T-value of the rFFA and an increase in performance or response time through the one-back repetition task.

Regarding the IEBA, there was no significant correlation between age and size at all thresholds. However, for the rEBA, size was negatively correlated with age at the thresholds of $p < .05$ and $p < .01$. The peak T-values of both the IEBA and rEBA demonstrated insignificant positive correlations with age. Although the rEBA was found to be smaller in adults than in children, selectivity did not decrease with age.

Concerning whole-brain group analysis, there was significant activation in the rFFA for the Faces > Tools category and significant activation in the rFBA, IEBA, and rEBA for the Bodies > Tools category in both age groups. Face-selective responses were weaker in children compared to adults, while body-selective responses were comparably strong across age groups.

Discussion

The present study investigated the developmental trajectories of face- and body-selective visual areas, and it has been found that the areas develop independently of one another. The rFFA increased in both size and selectivity with age, compared to the rFBA which showed no such development and remained comparable in size in both children and adults. Confounding factors, including the amount of head motion and variations in cortical anatomy, among others, have no influence on the results as these factors would equally affect the rFFA and the rFBA since these regions lie in the same part of the visual cortex. The one-back repetition task controlled for attention differences. One of the possibilities for the developmental path of the rFFA is that it may be directly related to an increase in perceptual processing of faces, continuing to develop through late childhood, creating category-selective responses in the fusiform gyrus.

For example, selectivity for language was found to have a positive correlation with exposure and experience to the language in the extrastriate cortex (Baker et al., 2007). In contrast, the lack of significant development of the rFBA after 7 years old may point towards the perceptual processing of bodies developing little beyond childhood. Another prospect for the independent developmental paths of the rFFA and the rFBA is that people are constantly exposed to new faces throughout their lifetimes. Thus, an increase in size and face selectivity would be necessary to discriminate and remember these new faces. Bodies, on the other hand, are not viewed as much due to clothing obscuring distinguishing body features. Furthermore, height may also influence face- and body-response activation. Since children are shorter than adults, they often focus on bodies more rather than faces, suggesting a consistent use of body-selective areas at a young age. Adults, on the other hand, focus more on the face as height increases, suggesting an increase in the use of face-selective areas with age. Since the rFBA is substantially larger than the rFFA in children than in adults, this insinuates that the rFBA develops earlier and/or faster than the rFFA.

In regards to the rFFA, the peak T-values in the children, adolescent, and adult groups suggest a gradual developmental increase in both size and selectivity.

Crediting the advanced ability of high-resolution fMRI, the FFA and FBA overlaps at around 54-75%. The development of the close proximity of these two regions is likely to help in combining face and body processing to allow for a holistic understanding of the emotions, identities, and intentions of others.

While the size of the rFFA showed a positive correlation with age and the size of the rFBA demonstrated to be relatively consistent with age, the developmental trajectory of the rEBA is opposite to that of the rFFA, displaying patterns of size decrease with an increase in age. Since it was determined that the rEBA displays a size reduction with age, there is a possibility that activation and focus of bodies shift from a holistic representation to a more focal representation.

In conclusion, although past research generally supported the view that various functions associated with a specific brain region tend to develop along similar trajectories due to the strong relationship between brain structure and functional development implies, the results of the present study demonstrate that this is not always the case.

Commentary

An understanding of the development and biological makeup of these face- and body-selective regions and how they interact can aid us in making progress in the clinical field of visual recognition defects, such as prosopagnosia.

Prosopagnosia is the inability to recognize familiar faces and can be brought on either by genetics — developmental prosopagnosia (DP) — or brain damage — acquired prosopagnosia. Developmental prosopagnosia is a rare, neurodevelopmental, heterogeneous disorder, meaning that its profile varies person by person, making it difficult to document and study the disorder (Susilo & Duchaine, 2013). One of the major damaged areas in prosopagnosia is the FFA, as it's strongly associated with facial identity processing (Grill-Spector et al., 2004). This visual disorder makes it difficult for those with prosopagnosia to nurture meaningful relationships so they opt to utilize non-facial cues such as body features to identify a person. However, since the FFA and FBA closely overlap, it is likely that lesions in the rFFA that cause prosopagnosia would also affect the FBA. This hypothesis can be supported by a study by Biotti et al. (2017) which found evidence that those with developmental prosopagnosia can also experience impaired body perception. Thus, it would be possible that those with prosopagnosia would also show impairment in body perception processing, making it even more challenging for people with prosopagnosia to lead a normative social life. Other studies found that in patients with developmental prosopagnosia, face- and body-selective mechanisms are less distinct (Van den Stock et al., 2008). In the FFA, there was lower activation for neutral faces and in the EBA, there was higher activation for neutral faces. This indicates

that, potentially, developmental prosopagnosia changes the developmental trajectory, and therefore mitigates the selectivity, of the two regions. That, along with findings by Scherf et al. (2007) that reveal category specialization is due to experience-dependent learning and the maturing brain may help explain the inability to improve prosopagnosia symptoms, especially in developmental prosopagnosia. So, it can be considered how prosopagnosia can affect the development of the fusiform gyrus as a whole. To add on, the present study has established that the development of the FFA, FBA, and EBA take independent paths, so there is the question of how prosopagnosia can affect the development of each region and by exploring it, we can possibly find answers that lead to a better quality of life for those with prosopagnosia. Building upon the present study, a new study can be conducted on prosopagnosia patients to track the development of the FFA and FBA. This can allow us to fill the blanks in understanding the degree of prosopagnosia, and just how intertwined, or not, face- and body-selective regions are when they lie in nearly identical spaces with similar functions.

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